

We say that Ψ is *uniform* if for every Σ and pair of TBoxes $\mathcal{T}, \mathcal{T}'$ with the same number of existentially quantified variables $\Psi(\mathcal{T}, \Sigma) = \Psi(\mathcal{T}', \Sigma)$. Let $z \in \{i, f, q, c\}$; then, Ψ is *z-admissible* if, for each $\mathcal{T}$ and Σ , $\mathcal{M}^{\Psi(\mathcal{T}, \Sigma)}$ is a $\equiv_{\Sigma}^z$ -module of $\mathcal{T}$. Finally, Ψ is *z-optimal* if $\mathcal{M}^{\Psi(\mathcal{T}, \Sigma)} \subseteq \mathcal{M}^{\Psi'(\mathcal{T}, \Sigma)}$ for every $\mathcal{T}, \Sigma$ and every uniform Ψ' that is *z-admissible*. $\diamond$

Uniformity ensures that settings do not depend on the specific shape of rules in $\mathcal{T}$, but rather only on Σ and the number of existentially quantified variables in $\mathcal{T}$. In turn, admissibility ensures that each setting yields a module. The (uniform and admissible) family Ψ^z corresponding to each setting χ^z in Sections 3.3 and 3.4 is defined in the obvious way: for each $\mathcal{T}$ and Σ , $\Psi^z(\mathcal{T}, \Sigma)$ is the setting χ^z for $\mathcal{T}$ and Σ .

The next theorem shows that Ψ^z is optimal for implication and classification inseparability.

Theorem 21. Ψ^z is *z-optimal* for $z \in \{i, c\}$.

In contrast, Ψ^q and Ψ^f are not optimal. To see this, let $\mathcal{T} = \{A(x) \rightarrow B(x), B(x) \rightarrow A(x)\}$ and $\Sigma = \{A\}$. The empty TBox is fact inseparable from $\mathcal{T}$ since the only Σ -consequence of $\mathcal{T}$ is the tautology $A(x) \rightarrow A(x)$. However, $\mathcal{M}^{\chi^f} = \mathcal{T}$ since fact $A(a)$ is in $\mathcal{A}_r^f$ and its support is included in the module. We can provide a family of settings that distinguishes tautological from non-tautological inferences (see appendix); however, this family yields settings of exponential size in $|\mathcal{T}|$, which is undesirable in practice.

4 Proof of Concept Evaluation

We have implemented a prototype system for module extraction that uses RDFS for datalog materialisation (?). Additionally, the ontology reasoner PAGODA (?) provides functionality for computing the support of an entailed fact in datalog, which we have adapted for computing modules. We have evaluated our system on representative ontologies, including SNOMED (SCT), Fly Anatomy (FLY), the Gene Ontology (GO) and BioModels (BM).¹ SCT is expressed in the EL profile of OWL 2, whereas FLY, GO and BM require expressive DLs (Horn-*SRI*, *SHIQ* and *SRIQ*, respectively). We have normalised all ontologies to make axioms equivalent to rules.

We compared the size of our modules with the locality-based modules computed using the OWL API. We have followed the experimental methodology from (?) where two kinds of signatures are considered: *genuine signatures* corresponding to the signature of individual axioms, and *random signatures* with a given probability for a symbol to be included. For each type of signature and ontology, we took a sample of 400 runs and averaged module sizes. For random signatures we considered a probability of 1/1000. All experiments have been performed on a server with two Intel Xeon E5-2643 processors and 90GB of allocated RAM, running RDFS on 16 threads.

Table 1 summarises our results. We compared $\perp$ -modules with the modules for χ_c (Section 3.4) and $\top\perp^*$ -modules with those for χ_m, χ_q, χ_f , and χ_i (Section 3.3). We can

¹The ontologies used in our tests are available for download at <http://www.cs.ox.ac.uk/isg/ontologies/UID/> under IDs 794 (FLY), 795 (SCT), 796 (GO) and 797 (BM).

	FLY		SCT		GO		BM	
rules	19,830		112,833		145,083		462,120	
	gen	rnd	gen	rnd	gen	rnd	gen	rnd
$\perp, \chi_b$	242	847	242	5,196	1,461	12,801	1,010	64,320
χ_c	112	446	230	3,500	309	3,990	285	14,273
$\top\perp^*$	219	796	233	5,182	1,437	12,747	963	62,897
χ_m	215	789	233	5,182	1,431	12,724	955	62,286
χ_q	109	480	123	2,329	267	4,146	447	16,905
χ_f	76	476	24	2,258	162	4,142	259	14,043
χ_i	8	7	15	235	103	2,429	105	4,107
$ \Sigma $	2.7	7.8	2.7	41.9	2.4	56.6	2.5	210.5

Table 1: Results for genuine and random signatures Σ

see that module size consistently decreases as we consider weaker inseparability relations. In particular, the modules for χ_c can be 4 times smaller than $\perp$ -modules. The difference between $\top\perp^*$ -modules and χ_i modules is even bigger, especially in the case of FLY. In fact, χ_i modules are sometimes empty, which is not surprising since two predicates in a large ontology are unlikely to be in an implication relationship. Also note that our modules for semantic inseparability slightly improve on $\top\perp^*$ -modules. Finally, recall that our modules may not be minimal for their inseparability relation. Since techniques for extracting minimal modules are available only for model inseparability, and for restricted languages, we could not assess how close our modules are to minimal ones and hence the quality of our approximation.

Computation times were comparable for all settings χ^z with times being slightly higher for χ_i and χ_c as they involved a larger number of constants. Furthermore, extraction times were comparable to locality-based modules for genuine signatures with average times of 0.5s for FLY, 0.9s for SCT, 4.2 for GO and 5s for BM.

5 Conclusion and Future Work

We have proposed a novel approach to module extraction by exploiting off-the-shelf datalog reasoners, which allows us to efficiently compute approximations of minimal modules for different inseparability relations. Our results open the door to significant improvements in common applications of modules, such as computation of justifications, modular and incremental reasoning and ontology reuse, which currently rely mostly on locality-based modules.

Our approach is novel, and we see many interesting open problems. For example, the issue of optimality requires further investigation. Furthermore, it would be interesting to integrate our extraction techniques in existing modular reasoners as well as in systems for justification extraction.

Acknowledgements

This work was supported by the Royal Society, the EPSRC projects MaSI³, Score! and DBOnto, and by the EU FP7 project OPTIQUE.

Itaque veritatis fugiat vitae doloremque nam quae excepturi earum autem dolore, nostrum voluptas veritatis exercitationem, quibusdam earum iure obcaecati fugiat, nostrum

pariatur perferendis illum blanditiis tempore similique quia
corporis hic. Expedita vel eos quod exercitationem consec-
tur repellat tempore beatae